

CENG381 Stochastic Processes

Renewal Processes and Elementary Renewal Theorem - Question Set 1

Notation used in this question set:

$X_1, X_2, \dots$	i.i.d. non-negative inter-renewal times with common CDF F , mean $E[X] = \mu$, and second moment $E[X^2]$.
$S_n = X_1 + \dots + X_n$:	the time of the n -th renewal ($S_0 = 0$ by convention). The renewals happen at times $S_1, S_2, S_3, \dots$
$N(t)$:	the number of renewals in the interval $(0, t]$. Equivalently, $N(t) = \max\{n \geq 0 : S_n \leq t\}$.
$m(t) = E[N(t)]$:	the renewal function. It counts the expected number of renewals by time t .
Elementary Renewal Thm:	$m(t) / t \rightarrow 1 / \mu$ as $t \rightarrow \infty$. The long-run rate of renewals equals $1 / (\text{mean inter-renewal time})$.

Key identity (used to derive $m(t)$): $E[N(t)] = \sum_{n=1}^{\infty} P(S_n \leq t) = \sum_{n=1}^{\infty} F_n(t)$, where F_n is the n -fold convolution of F with itself.

Q1 (30 points). Light bulbs in a lamp are replaced immediately when they fail. Each bulb has a lifetime that is Exponential with rate $\lambda = 0.5$ failures per hour (so $E[X] = 1/\lambda = 2$ hours).

- What type of stochastic process does $N(t)$ (the number of replacements by time t) form? Write the distribution of $N(t)$ and compute $E[N(t)]$.
- Use the Elementary Renewal Theorem to find the long-run average rate of replacements per hour. How many bulbs do you expect to use in a 40-hour period?
- The first bulb is installed at time 0. Find the probability that at least 3 replacements occur in the first 4 hours. Since $N(t) \sim \text{Poisson}(\lambda t)$, compute $P(N(4) \geq 3)$ using the Poisson PMF with $\lambda t = 0.5 * 4 = 2$.

Q2 (35 points). Buses arrive at a stop according to a renewal process. Inter-arrival times X_n are uniformly distributed on $[1, 3]$ minutes (so $E[X] = 2$ minutes, $E[X^2] = \int_1^3 x^2 * (1/2) dx$ from 1 to 3).

- Compute $E[X]$ and $E[X^2]$ for $X \sim \text{Uniform}[1, 3]$. Recall: for $X \sim \text{Uniform}[a, b]$, $E[X] = (a+b)/2$ and $E[X^2] = (a^2 + ab + b^2)/3$.
- By the Elementary Renewal Theorem, what is the long-run rate of bus arrivals per minute? How many buses arrive on average in a 60-minute hour?
- The renewal equation gives the exact renewal function for small t . For a $\text{Uniform}[1, 3]$

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inter-arrival distribution, $N(t) = 0$ for $t < 1$ (the first bus cannot arrive before minute 1). For $1 \leq t < 2$, at most one bus has arrived, so $m(t) = E[N(t)] = P(X_1 \leq t)$. Compute $m(1.5) = P(X_1 \leq 1.5)$ for $X_1 \sim \text{Uniform}[1,3]$.

Q3 (35 points). A web server processes requests in batches. The time to process each batch X_n has the following distribution:

$$P(X = 1) = 0.5, \quad P(X = 2) = 0.3, \quad P(X = 4) = 0.2$$

So inter-renewal times take values 1, 2, or 4 seconds.

- a) Compute $E[X]$ and $E[X^2]$ for this distribution.
- b) What is the long-run rate of batch completions per second (by the Elementary Renewal Theorem)?
- c) Use the identity $P(N(t) \geq n) = P(S_n \leq t)$ to compute:
 - (i) $P(N(3) \geq 1) = P(S_1 \leq 3) = P(X_1 \leq 3) = P(X=1) + P(X=2)$
 - (ii) $P(N(3) \geq 2) = P(S_2 \leq 3) = P(X_1 + X_2 \leq 3)$.

For part (ii), enumerate all combinations of (X_1, X_2) that both take values in $\{1, 2, 4\}$ and sum to at most 3.